Every Move I Make:
The Relationship Between Metacognition and Motor Performance in Guitar Playing

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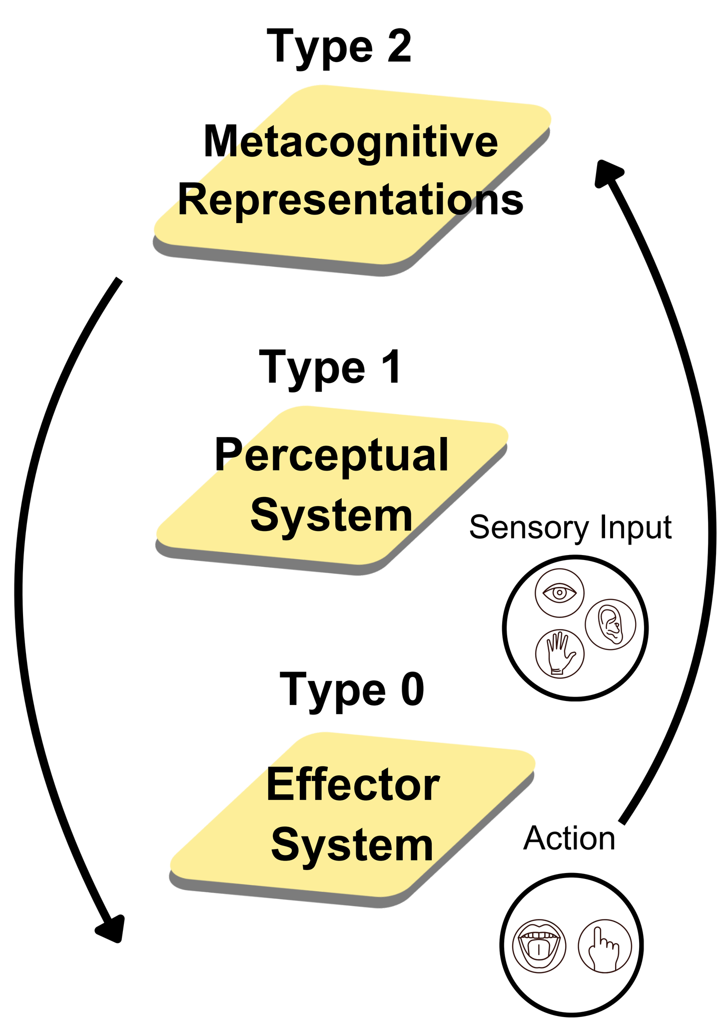
Background

Metacognition and Motor Performance

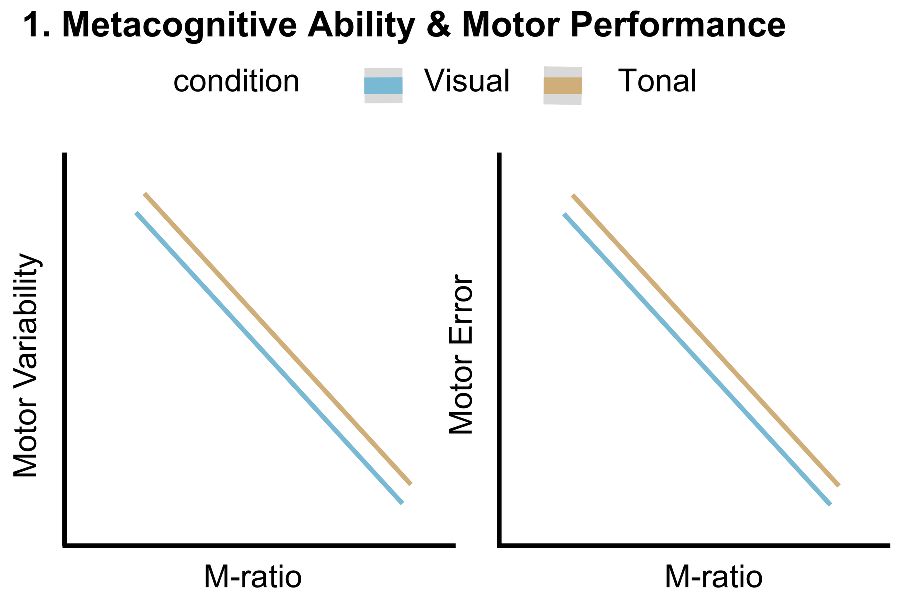
1. It is often assumed that **metacognition** improves **motor performance** (Concina, 2019; MacIntyre et al., 2014), but this relationship is seldom tested. **Does it?**

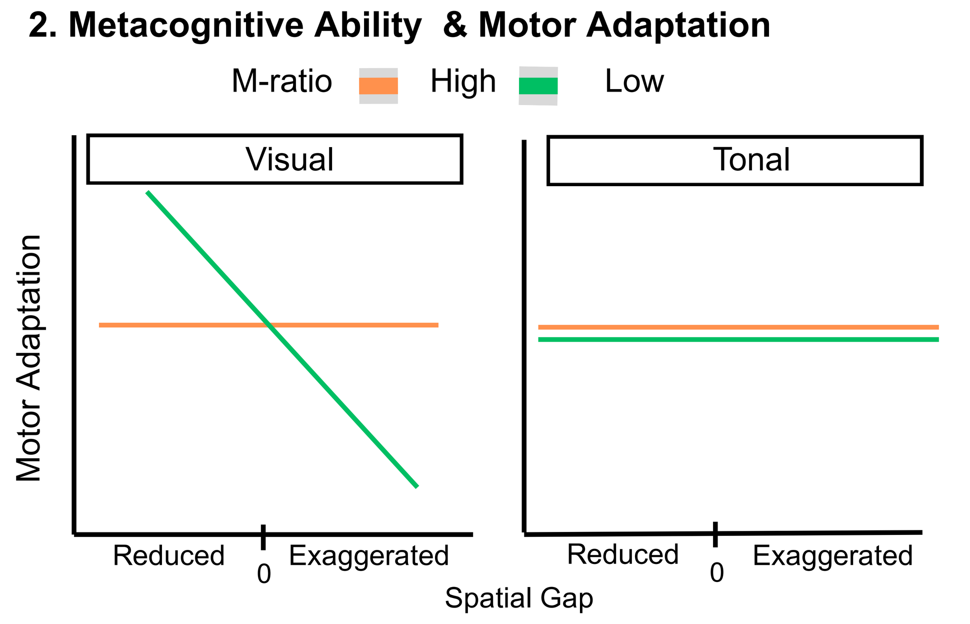


2. Humans can **adapt their movements** when there are **changes** in the environment or the body, often without being aware of them (Blakemore et al., 2002; Fourneret & Jeannerod, 1998; Shadmehr & Mussa-Ivaldi, 1994). Does the degree of **motor adaptation** change with the precision of **metacognitive representations** of one's own movements?

(Shea et al., 2014)

Hypotheses



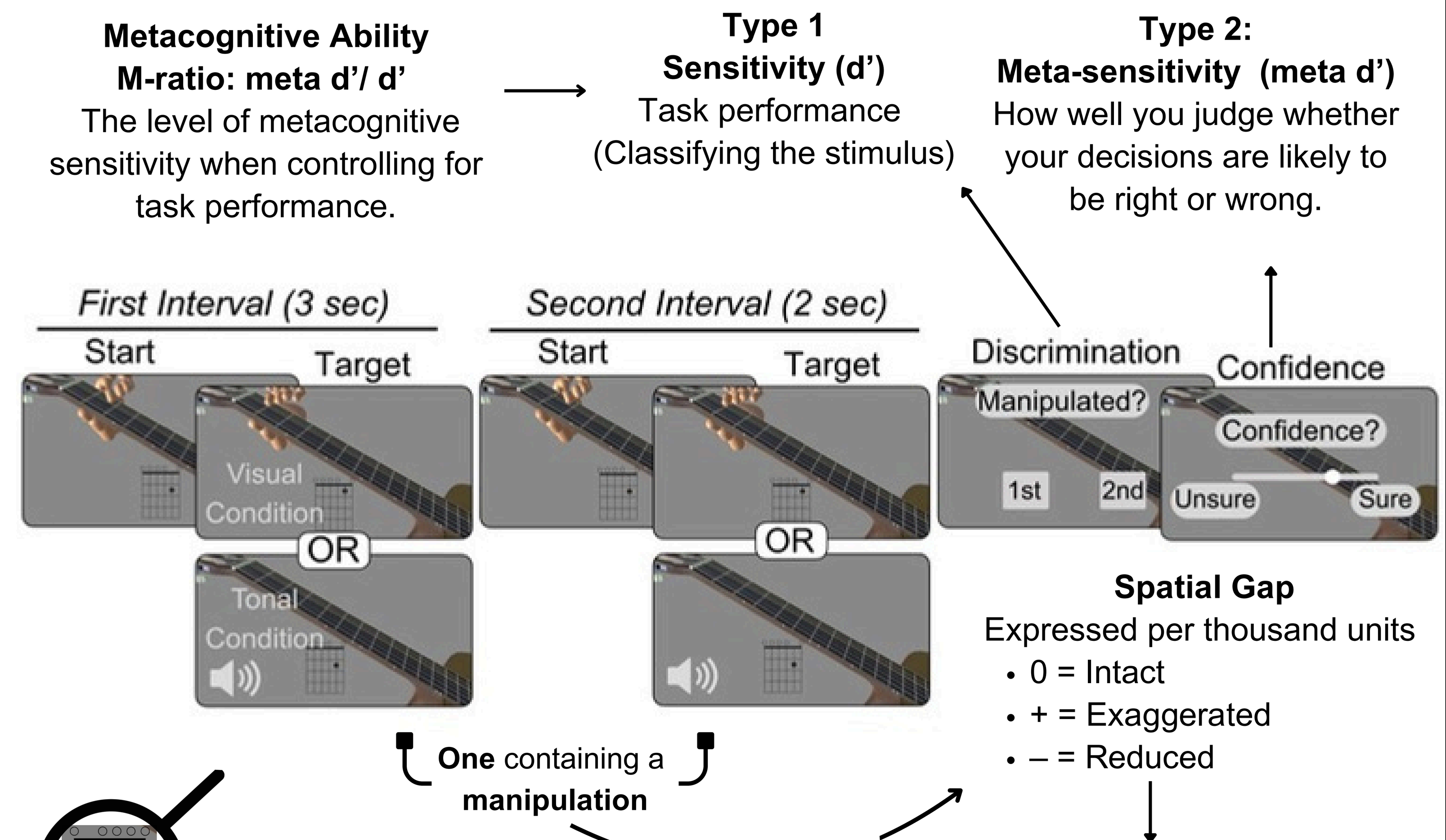


Methods

Secondary Data Analyses: 65 amateur guitarists (44 after exclusions: M = 26.07 years, SD = 5.93).

See the pre-print for full methods

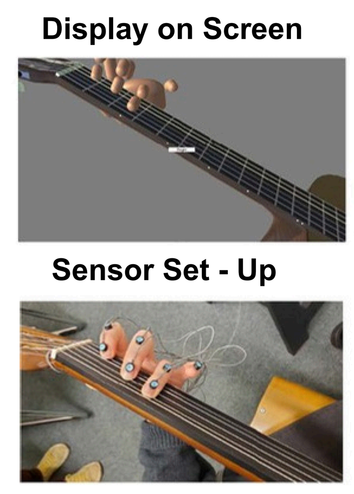
Metacognitive Guitar Task



Spatial Gap
Expressed per thousand units
- 0 = Intact
- + = Exaggerated
- – = Reduced

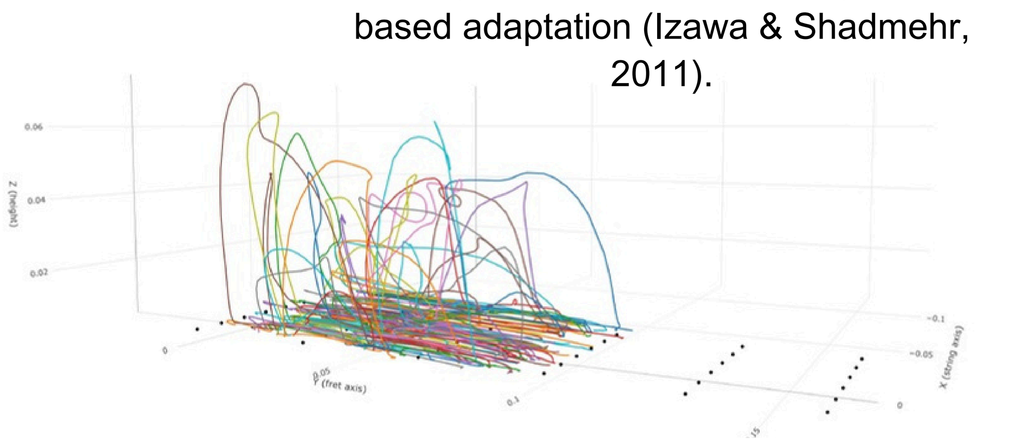
Type 0
Execution of the motor task itself
(Playing the note correctly)

Motor Adaptation
Comparison of **visual** intervals with vs. without manipulation.
Note: Tonal feedback was **endpoint-only** and provided **no perceptible change** during movement, **preventing** error-based adaptation (Izawa & Shadmehr, 2011).



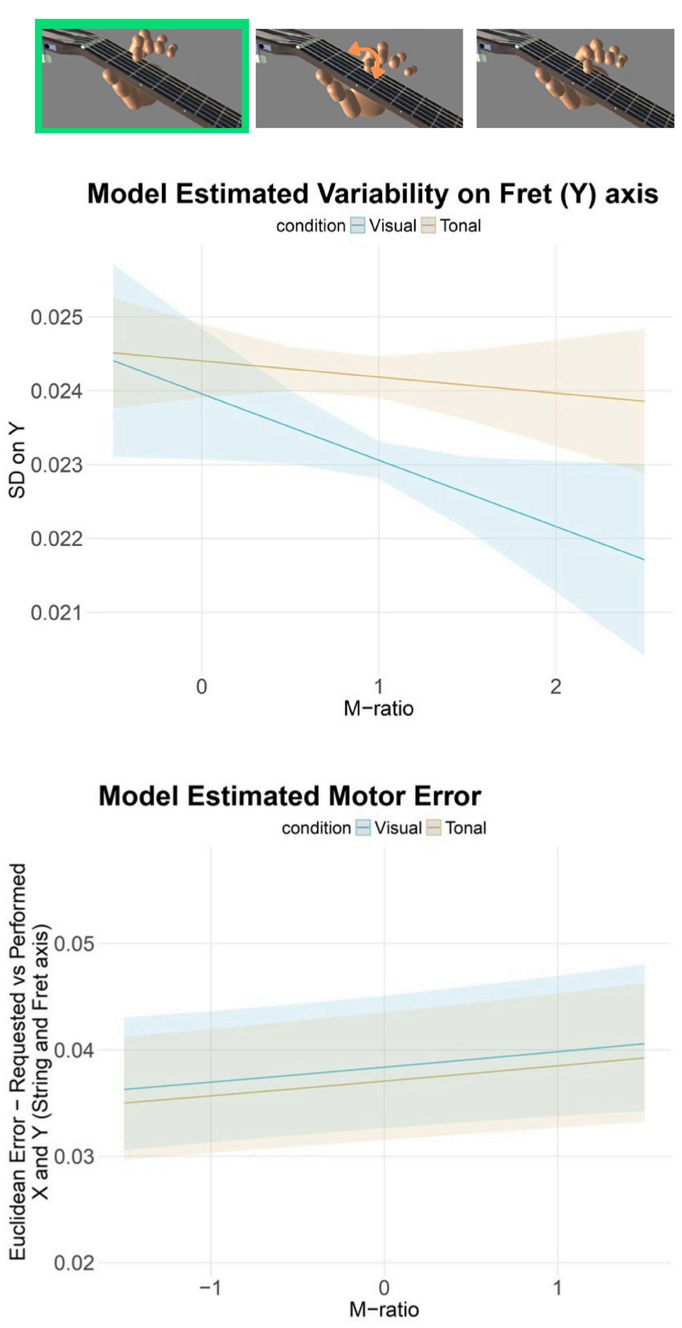
(Fretless Guitar)

Positional data (x, y, z)

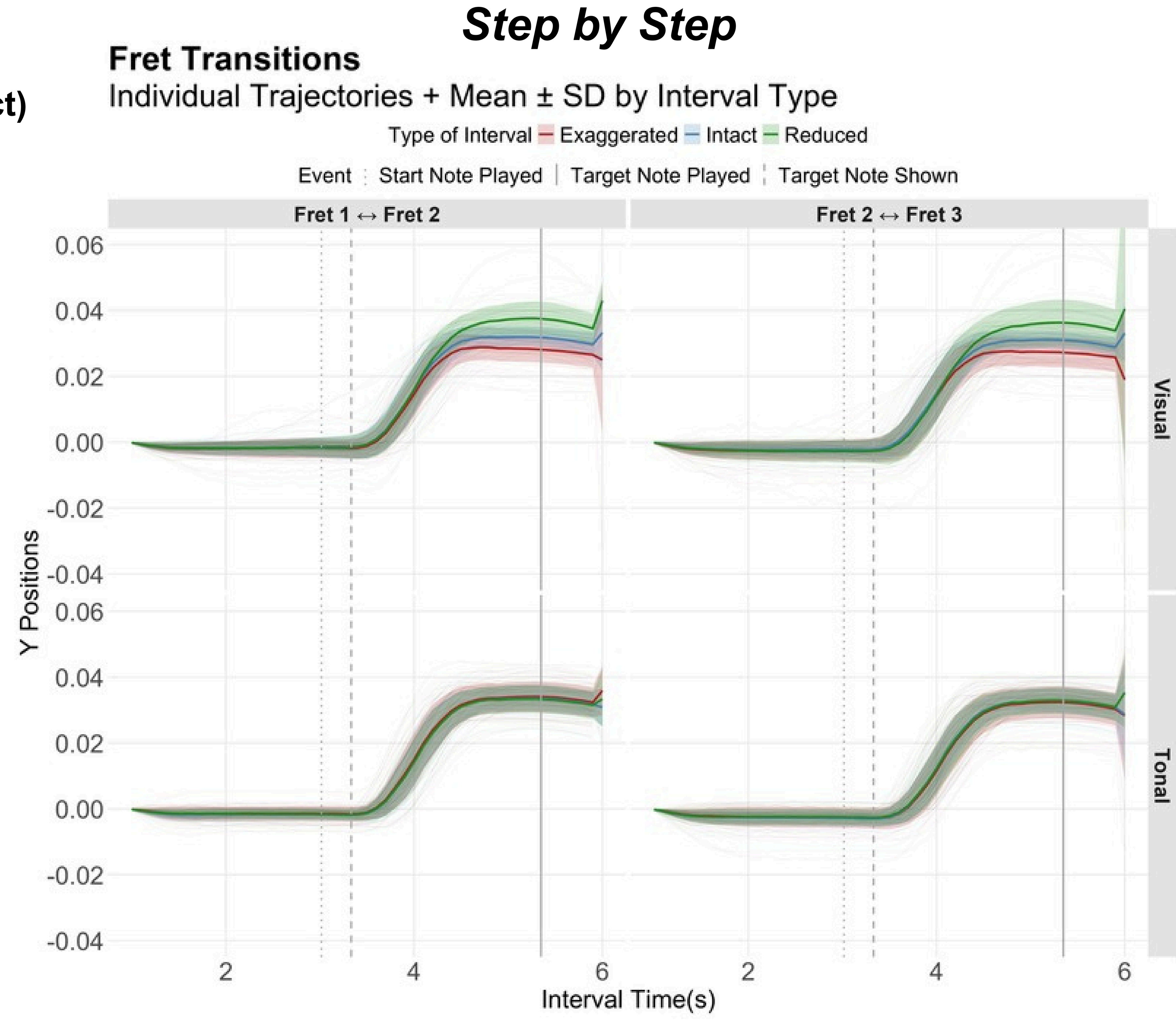


Results

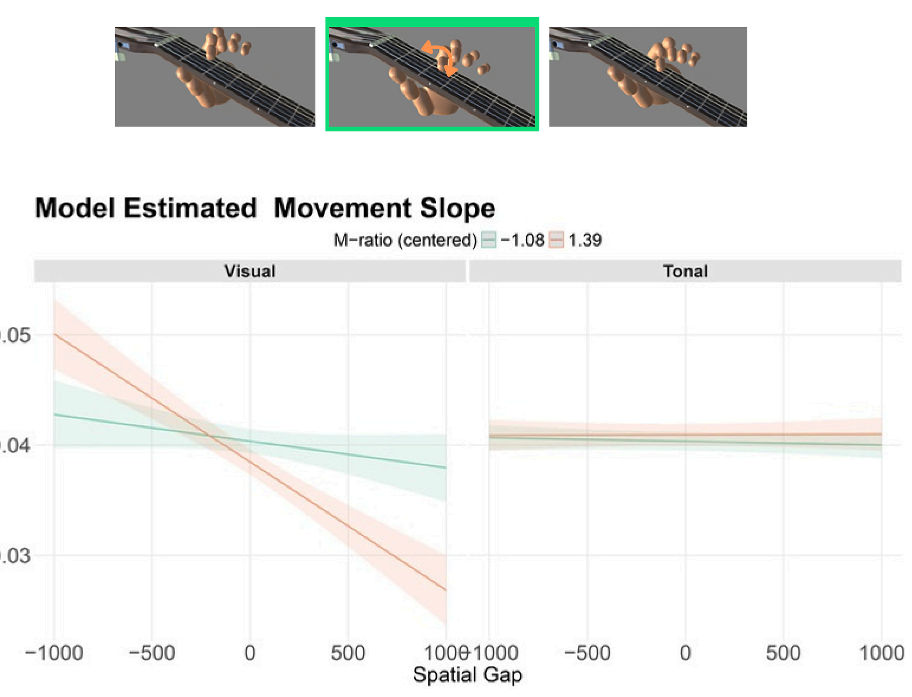
Note 1 (Start Position - Intact)



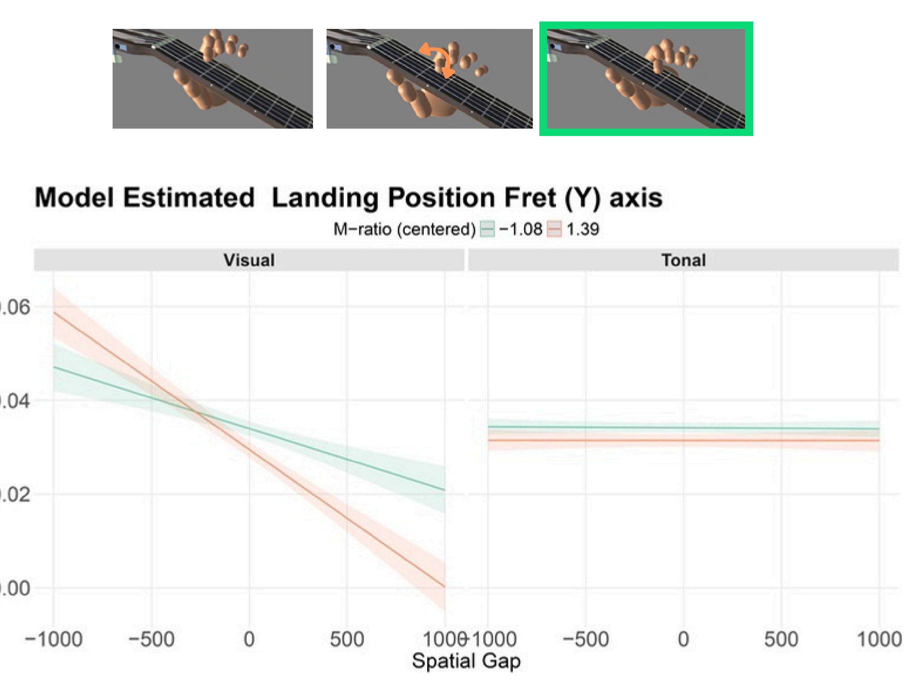
Step by Step



Trajectory (Slope)



Note 2 (Target Position)



Discussion

Metacognitive ability & motor performance
- Higher metacognitive ability → lower movement variability, but not greater accuracy (landing error). However, our target measures (x and y coordinates of fret and string) may not fully capture musical performance accuracy.
- This precision–accuracy dissociation aligns with prior work on motor variability (van Beers et al., 2004; Müller & Sternad, 2004).
- High M-ratio participants may consciously produce more consistent movements (Trempe et al., 2011)

Metacognitive ability & motor adaptation
- Participants, on average, adapted to the manipulation, though extent varied by metacognitive ability and condition.
- High m-ratio participants showed greater adaptation (steeper trajectories/landings).
  - Greater explicit (fast/larger) vs implicit (slow/smaller) adaptation in high vs low m-ratio (McDougle, 2015).
  - High m-ratio offload suppressing adaptation to prioritize metacognitive monitoring (Risko & Gilbert, 2016).

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